

Brandora: AI-Powered Brand Identity Generator

FIELD TRAINING PROJECT REPORT

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ABSTRACT

Brandora is an end-to-end AI system that generates a complete visual brand identity from a short textual brief. Given an industry, target audience, personality, tone, and purpose, the system produces three brand proposals (name, slogan, color palette, typography direction, and a concrete logo concept). After proposal selection, a logo is generated and exported as a downloadable Brand Kit PDF. Driven by Qwen3-8B and FLUX.1-schnell (4-bit quantization on a single 16 GB GPU), CLIP evaluation identified the correct brand for 57% to 62% of 21 logos (12× above chance) with a 24% reduction in color deviation via palette enforcement.

1. INTRODUCTION AND OBJECTIVES

Creating a brand identity normally requires a designer and several iterations. The goal of this project was to demonstrate an end-to-end AI system combining text and image generation using fully free resources, supported by a documented model-comparison methodology.

- **Structured Generation:** Reliably produce brand specifications from free-text briefs.
- **Concept-Driven Logos:** Generate logos reflecting the concept and exact color palette.
- **Interactive Workflow:** Interface for users to compare proposals and export a Brand Kit PDF.
- **Quantitative Evaluation:** Benchmark visual alignment using CLIP ViT-B/32 metrics.

2. SYSTEM ARCHITECTURE

The Brandora processing pipeline consists of five sequential stages designed for resource-constrained execution:

1. **Brief Input (Gradio):** Accepts industry, target audience, purpose, traits, and tone.
2. **Proposal Generation (Qwen3-8B):** Produces 3 distinct brand packages as validated JSON with built-in diversity filtering (rejects generic words, duplicate names, and prefix collisions).
3. **Selection Interface:** Interactive cards displaying brand name, tagline, and swatches.
4. **Logo Generation (FLUX.1-schnell):** Synthesizes logo from concept/shape language; followed by post-generation pixel recoloring to the exact palette.
5. **Brand Kit Export:** Assembles high-resolution assets into a structured PDF document.
6. **Project Theme:**

Deep Forest #0D1F16	Pine Shade #14381F	Emerald Depth #1C4D2E	Moss Green #2E6B3F	Botanical Green #3F8451	Soft Fern #5AA06D	Sage Leaf #7BAE8A	Olive Mist #A1C29F	Linden Green #C3DDBA	Pistachio Cream #E6F2E1
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3. MODEL SELECTION & INFRASTRUCTURE

Models were selected based on license, speed, and adherence to structured JSON output under 4-bit quantization:

Task	Selected Model	Technical Justification & Notes
Text Generation	Qwen3-8B (4-bit NF4)	Apache 2.0; consistently produced valid JSON across 30 standardized briefs.
Logo Generation	FLUX.1-schnell (4-bit transformer/T5)	Fast execution (4 steps); open weights; CPU offload enabled.

**Rejected models:* LiquidAI/LFM2.5-2.6B (inconsistent JSON), SD3 Medium (compatibility), Qwen-Image-2.1 & PlaygroundV2.5 (evaluated but unselected).

4. ENGINEERING CHALLENGES AND SOLUTIONS

Problem	Root Cause	Engineering Solution
CUDA Out-Of-Memory	Both models competing for 16 GB GPU; VAE requested 5.5 GB.	Sequential execution: explicit unloading of Qwen before FLUX runs; VAE tiling/slicing; expandable segments; resolution fallback to 512px. GPU memory verified at ~0.01 GB post-unload.
Export Crash	Gradio passed non-PIL image formats.	Added robust conversion handling PIL, NumPy, path, tuple, and dict inputs; set type="pil".
Prompt Truncation	CLIP branch limited to 77 tokens.	Dual-prompting: compact prompt for CLIP and comprehensive prompt for T5 (prompt_2).
Generic Logo Outputs	Style description preferred over symbol; weak negative prompts.	Qwen generates explicit logo_concept & logo_shape_language; positive-only prompting with seed randomization.
Palette Deviation	FLUX ignores strict RGB color instructions.	Post-generation palette enforcement: mapping pixels to nearest palette color (+ white background).
UI Readability Issues	Theme color collisions in dark mode.	Custom CSS variables enforcing brand greens and fixed text contrast.
Race Conditions	Repeated clicks on generation button.	Button disabled state handling with concurrency limit set to 1.

5. QUANTITATIVE EVALUATION METHODOLOGY

A sample of 21 standardized briefs across diverse industries (e.g., streetwear, meditation, law firm, perfume, bakery) was evaluated in two pipeline states: raw FLUX output and post-palette enforcement.

CLIP ViT-B/32 METRICS DEFINITION

- Concept Score:** Cosine similarity between logo image and text *"a minimalist logo of [logo_concept]"*.
- Margin:** Concept score minus average score against all other 20 brand concepts (positive = correct targeting).
- Retrieval Accuracy:** Fraction of logos correctly matched to their concept out of 21 candidates (Random chance = 0.048).
- Palette Distance:** Mean RGB distance between downsampled logo pixels (64×64) and nearest palette color.

6. EVALUATION RESULTS

61.9% RETRIEVAL ACCURACY	13× ABOVE CHANCE	-23.7% PALETTE DISTANCE
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Pipeline Version	Concept Score	Margin	Retrieval Acc.	Text-Free Score	Palette Dist.
FLUX Raw Output	0.306	+0.075	0.571 (12/21)	0.803	11.4
After Palette Enforcement	0.303	+0.067	0.619 (13/21)	0.797	8.7

Key Findings: Positive margins (+0.075 / +0.067) and high retrieval accuracy verify strong semantic binding between generated text concepts and visual symbols. Post-processing reduced color deviation by ~24% (11.4 → 8.7) with negligible impact on semantic concept score (0.306 → 0.303).

7. LIMITATIONS

- **Semantic Focus:** CLIP evaluates concept alignment, not aesthetic design quality or human preference.
- **Sample Size:** 21 logos provide a preliminary benchmark; minor percentage variations are non-conclusive.
- **Text Detection:** Zero-shot text detection in logos via CLIP ViT-B/32 remains approximate.
- **Sequential Latency:** Model loading/unloading on a single GPU adds runtime overhead.

8. FUTURE WORK

- Human evaluation study employing structured designer rubrics.
- Deployment on Hugging Face Spaces (ZeroGPU) with persistent model memory residency.
- Direct SVG vector graphic generation and export.

9. CONCLUSION

Brandora demonstrates a practical end-to-end framework for automated brand identity generation on modest, single-GPU infrastructure. By combining sequential memory management, dual-prompting, and post-generation palette enforcement, the system achieves strong visual concept fidelity significantly exceeding random baseline performance.

APPENDIX A. EVALUATION TOOLS: CLIP AND ALTERNATIVES

This appendix explains how the CLIP metric used in Section 5 works, why it was selected, and how it compares with other evaluation tools. **Only CLIP ViT-B/32 was used in this project**; the alternatives were reviewed but not tested.

HOW CLIP ViT-B/32 WORKS

- **Two encoders.** The image encoder (a Vision Transformer) resizes the image to 224×224, splits it into 49 patches of 32×32 pixels, and passes them through 12 Transformer layers to produce a 512-dimensional vector. The text encoder (a Transformer limited to 77 tokens) maps a sentence to a vector of the same size.
- **Similarity.** Both vectors are L2-normalized, so their dot product equals their cosine similarity, which measures how well the image matches the text.
- **Training.** CLIP was trained on about 400 million image-text pairs with a contrastive cross-entropy loss over an N×N similarity matrix, which pulls matching pairs together and pushes mismatched pairs apart. Here the weights are frozen, so no training or loss computation occurs.
- **Use in Brandora.** (1) Concept score; (2) retrieval accuracy and margin, i.e., whether a logo is closer to its own concept than to other brands' concepts; (3) text-free score, a zero-shot comparison of “a logo with no text or letters” against “a logo with text and letters”.

Why CLIP was selected

CLIP is lightweight and runs on CPU, so it does not compete with FLUX for GPU memory. It is fast, widely used as a standard measure of text-image alignment, fully reproducible on free resources, and requires no human raters for the primary metrics.

Known limitations

- It measures semantic alignment only, not aesthetics, originality, or professional quality.
- It is weaker at compositional relations and negation, so the text-free score is approximate.
- Images are downsampled to 224×224, so fine details are lost.
- Raw similarity values (typically 0.2 to 0.35) have no absolute meaning; they are interpreted relative to a baseline (random chance and other brands' concepts).
- Each logo is compared with the same concept text that guided its generation, so the metric reflects fidelity to the concept, not the quality of the concept itself.

Comparison with alternatives

Method	Measures	Strengths	Weaknesses	Cost	Role in this project
CLIP ViT-B/32	Text-image alignment	Lightweight, fast, standard, reproducible	Weaker on composition and negation; does not measure aesthetics	CPU	Primary metric (used)
OpenCLIP (e.g., ViT-L/14)	Text-image alignment	Same approach with a larger model; generally more accurate	Slower and heavier	CPU / GPU	Secondary check
SigLIP 2	Text-image alignment	Often more accurate at similar size (sigmoid loss)	Requires small code changes	CPU / GPU	Secondary check
BLIP (ITM head)	Fine-grained text-image matching	Models relations between image regions and words	Slower; shorter text limit	GPU preferred	Possible
Aesthetic predictor / ImageReward	Visual appeal and human preference	Covers CLIP's main blind spot (quality)	Not trained specifically on logos	CPU / GPU	Complementary (future work)

Role in the evaluation and future work

CLIP is the primary automated metric because it directly answers whether a logo reflects its brand concept. Since none of these models judge creativity or design quality, automated scores should be complemented by human evaluation (see evaluation/rubric.md). Planned improvements:

- **ImageReward** (a model trained on human preference judgments of text-to-image outputs) can be added as a complementary metric that estimates perceived quality, addressing CLIP's inability to assess aesthetics. Because it was not trained specifically on logos, its scores should be validated against human ratings before being relied on.